Playback-to-microphone delay

Audient iD14 · Edifier speaker · Earthworks M50

Run 1 median

38.250 ms

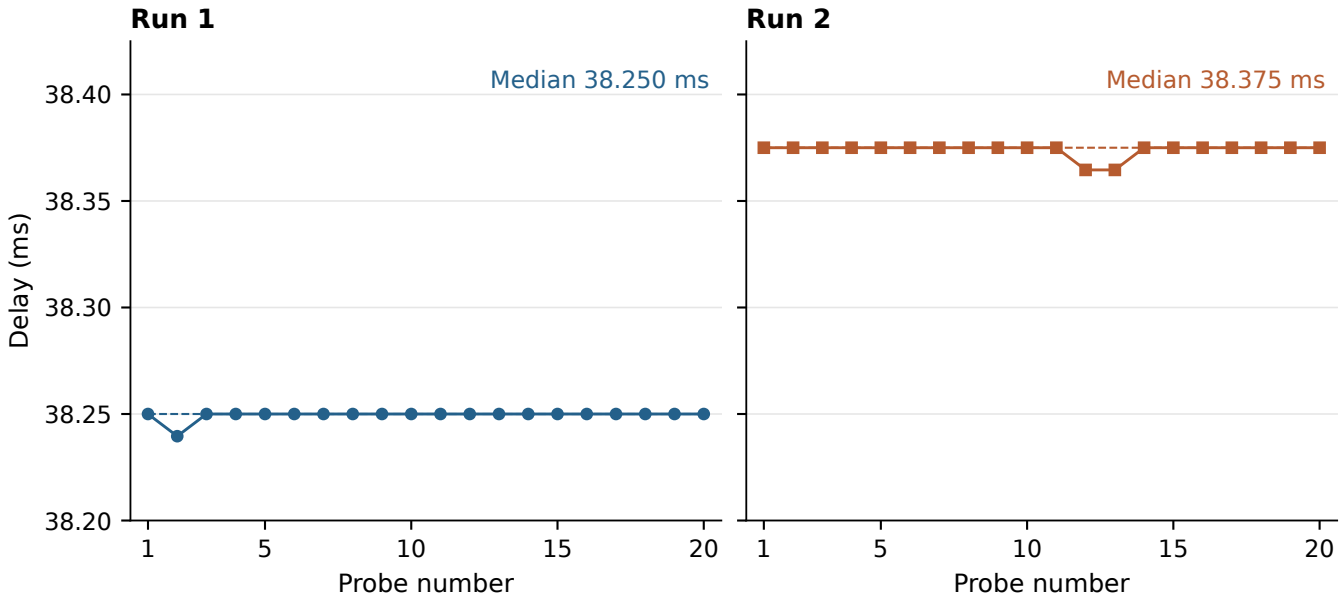
Sample SD ($n-1$): **0.00233 ms**

Run 2 median

38.375 ms

Sample SD ($n-1$): **0.00321 ms**

Recorded at 96 kHz, with approximately 10 cm microphone separation. Each separately opened stream provided 20 accepted sweeps.



Ranges: Run 1, 38.2396–38.2500 ms; Run 2, 38.3646–38.3750 ms. Each run's peak-to-peak spread was one sample (0.0104 ms). Probes were 100 Hz–20 kHz sweeps.

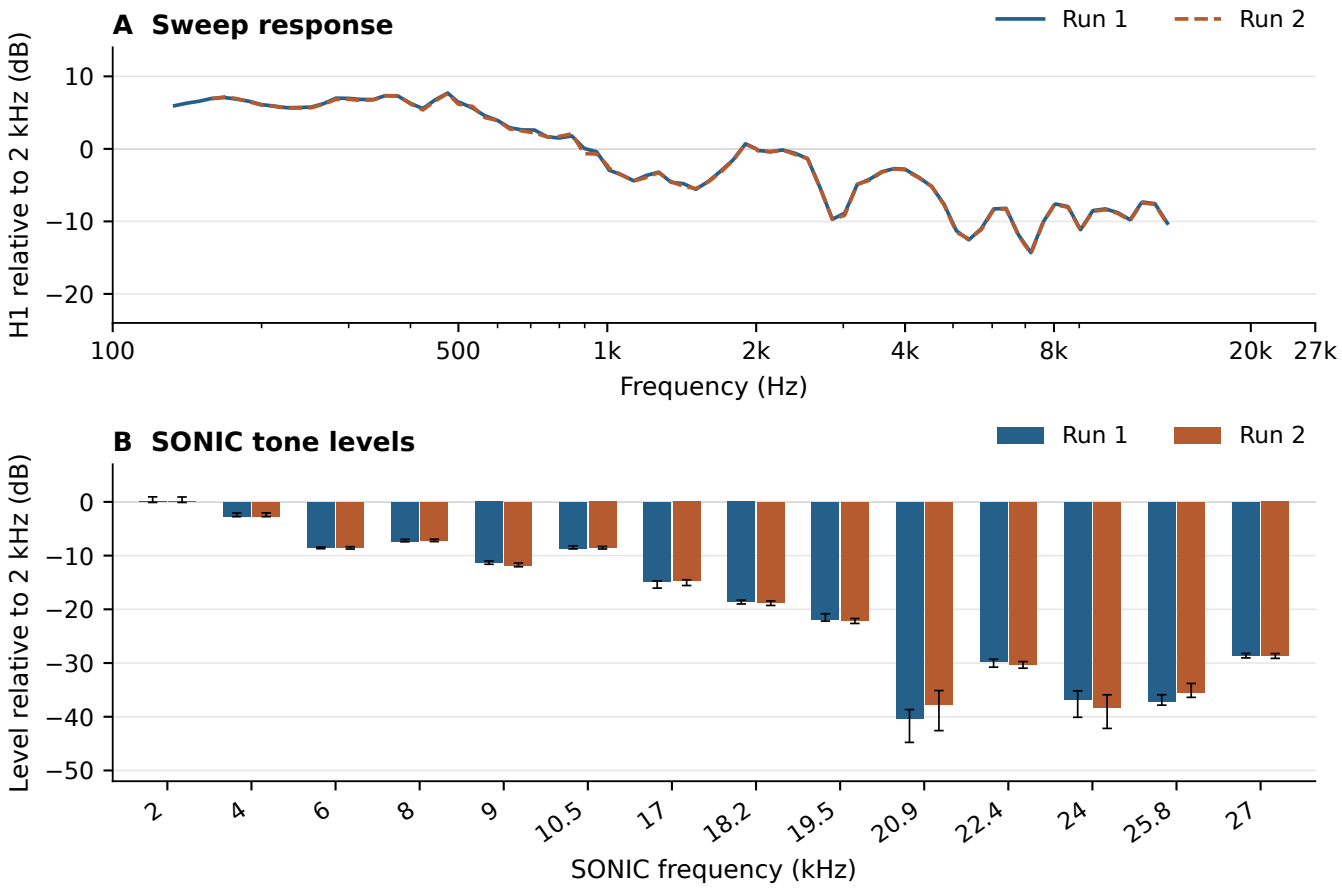
Description. The run medians differed by 0.125 ms. Sample SD describes variation within each run's 20 quantized waveform estimates. These application roundtrip measurements include buffering and the acoustic path; the SD values do not establish calibrated timing accuracy.

Recorded frequency levels

SONIC tone reconstruction · 14 frequencies · 2–27 kHz

Tone duration	Ramp duration	Recorded pips
10 ms	**2 ms**	**16,800**

Locally reconstructed Sheep E tones in two 84-second blocks.



Gaps mark unsupported sweep bands. Tone bars show medians; whiskers are 5th–95th percentiles of 600 pips per frequency per recording, not confidence intervals.

Description. All 14 components appear in both recordings at unequal levels. Relative to 2 kHz, 17–19.5 kHz was 15–22 dB lower; 20.9–25.8 kHz was 30–40 dB lower. These uncorrected full-chain measurements leave SPL, speaker-only response and distortion unqualified.

Next. Fix the geometry, calibrate microphone/SPL, test separated tones, and validate timing with an independent electrical reference.